

Spontaneous $\mathbb{Z}_2$ Symmetry Breaking via Hysterical Instability

A master branch-asymmetry theorem and a catalogue of odd channels

Andrew Bruce Boyd

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Computational note. Proofs, exploratory experiments, and paper writing were assisted by generative AI. Mathematical theorems stated in this paper are formally verified in Lean 4 in the companion specifications, as faithful ports of the TeX arguments at the abstractions recorded there, except results explicitly tagged as *literature hypotheses* (standard theorems cited from the literature and not re-proved here) or *physical inputs* (modeling assumptions outside mathematics).

Abstract

The electroweak baryon and chiral $B-L$ asymmetries are specializations of one pattern: a supercritical odd Hysterical collective mode spontaneously breaks a conjugation $\mathbb{Z}_2$, and a concurrent odd portal converts that breaking into a branch-conditioned charge asymmetry. This paper isolates the *generic* theorem, independent of the Standard Model channel, and then catalogues interesting $\mathbb{Z}_2$ symmetries in nature under a common five-column template (symmetry / odd coordinate / neutral portal / early-universe activity / broken in nature?). We do *not* claim that Hysterical Response caused any particular observed asymmetry. We do record an interpretive speculation: the chiral Standard Model Lagrangian may be the branch-selected remnant of a more symmetric parent theory, with weak parity appearing “already broken” because our classical patch sits on one Hysterical branch.

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1 Scope and status

What is proved.

$$\boxed{\begin{array}{l} \text{supercritical odd Hysterical mode + odd portal + nonzero overlap} \\ \implies \eta[Y_+] = -\eta[Y_-] \neq 0. \end{array}} \quad (1)$$

Here η is any washout-weighted yield sourced by an odd density $\Sigma[Y]$ with $\Sigma[-Y] = -\Sigma[Y]$. Criticality at *some* cosmic scale in the late-universe dilution class is already guaranteed [3]; the present paper does not re-prove it.

What is not proved. We do not prove that any particular cosmological asymmetry occurred, that the dilution scale a_c overlapped any given portal window, or that the Standard Model descended from a $\mathbb{Z}_2$ -symmetric parent Lagrangian. The catalogue in Section 5 records structural candidates and observational status; it is not a derivation of Nature’s history. The weak-parity discussion in Section 6 is labelled speculation.

Item	Status
Master branch-asymmetry implication (1)	Theorem 4.4 (Lean).
Odd $\mathbb{Z}_2$ pitchfork $Y_{\pm}$	Inherited [2]; recalled here.
Reciprocal destabilisation criterion	Theorem 3.2 (Lean).
Eventual dilution criticality at some a_c	Late-universe package [3].
Channel tables / portal identifications	Physical inputs / literature status.
“Hysterical Response caused X in Nature”	Not claimed for any row.
Weak-parity parent-Lagrangian reading	Speculation (Section 6).

2 Inherited odd response structure

The response-theory paper [2] defines

$$F_H = -\text{Tr}(F\mathcal{L}^{-1}\mathcal{S}) \quad (2)$$

on a stable response subspace, and isolates an exact conjugation symmetry

$$(Y, h) \mapsto (-Y, -h) \quad (3)$$

under which the local constitutive maps are odd. Near neutrality the closed-loop linearisation is $J = -\Gamma(I - K)$; in the reciprocal class, neutrality is unstable when $\lambda_{\max}(K) > 1$. A generic supercritical scalar normal form is

$$\dot{Y} = r_H Y - u Y^3 + O(Y^5), \quad u > 0, \ r_H > 0, \quad (4)$$

with saturated branches

$$\boxed{Y_{\pm} = \pm \sqrt{\frac{r_H}{u}} + O(r_H^{3/2}).} \quad (5)$$

Thus supercriticality spontaneously breaks the odd $\mathbb{Z}_2$ (3): perturbations along the unstable mode select a definite hand Y_+ or Y_- .

Definition 2.1 (Neutral portal). An effective coupling

$$\mathcal{L}_{\text{int}} = \frac{c_H}{f_H} Y \mathcal{O}, \quad \frac{c_H}{f_H} \neq 0, \quad (6)$$

is a *neutral portal* for the conjugation (3) when the operator $\mathcal{O}$ is odd under the same $\mathbb{Z}_2$ (so that (6) is even as a Lagrangian density) and is available already in the constitutive sector used to linearise about $Y = h = 0$.

The name records that the portal is part of the near-neutral odd response structure, not that $\mathcal{O}$ is electrically neutral.

3 Reciprocal destabilisation

Definition 3.1 (Two-mode reciprocal quadratic form). For $k_X, k_Y > 0$ and $g \in \mathbb{R}$,

$$F_2(X, Y) = \frac{1}{2} k_X X^2 + \frac{1}{2} k_Y Y^2 - gXY. \quad (7)$$

Theorem 3.2 (Reciprocal Hysterical Destabilisation). *Assume $k_X, k_Y > 0$ and $g \neq 0$. The neutral point $(X, Y) = (0, 0)$ of 3.1 is linearly asymptotically stable, critical, or unstable according as $g^2 \leq k_X k_Y$. Equivalently, instability holds precisely when $|g| > \sqrt{k_X k_Y}$.*

Proof. The Hessian determinant is $k_X k_Y - g^2$. With $k_X > 0$, positive-definiteness is equivalent to a positive determinant; a negative determinant yields eigenvalues of opposite sign, hence instability of gradient (or same-linearisation) dynamics. $\square$

4 Master branch-asymmetry theorem

Assumption 4.1 (Odd source). On a fixed even background (temperature, entropy density, washout rate, ...), a charge-density source functional obeys

$$\Sigma[-Y] = -\Sigma[Y]. \quad (8)$$

Assumption 4.2 (Weak-bias kernel form). In the regime used below,

$$\Sigma[Y](t) = \kappa K(t) \dot{Y}(t), \quad \kappa \neq 0, \quad K(t) \geq 0, \quad (9)$$

where K is an activity kernel for the portal's processing channel (sphaleron rate, chiral-gravity kernel, topological susceptibility, ...).

Assumption 4.3 (Washout-weighted transport). The map from source history to final yield is

$$\eta[Y] = \Phi(\Sigma[Y]), \quad (10)$$

where Φ is odd, sends nonnegative sources to nonnegative yields, and is strictly positive on any nonnegative source that is positive on a nonempty set.

The integrating-factor solution of $\dot{\eta} + \Gamma_W \eta = \Sigma$ with $\eta(t_i) = 0$ realises such a Φ whenever the kernel is nontrivial; Lean ports the abstract axioms rather than a particular ODE solver.

Theorem 4.4 (Master Hysterical branch asymmetry). *Assume 4.1–4.3 with $\kappa > 0$, and let $Y_-(t) = -Y_+(t)$ on a common even background. If the overlap condition*

$$\exists t : K(t) > 0 \text{ and } \dot{Y}_+(t) > 0 \quad (11)$$

holds (or the same with both signs flipped), then

$$\boxed{\eta[Y_+] = -\eta[Y_-] \neq 0.} \quad (12)$$

Proof. Under (9), $\Sigma[Y_-] = -\Sigma[Y_+]$. Oddness of Φ gives $\eta[Y_-] = -\eta[Y_+]$. Nonnegativity of K and of $\dot{Y}_+$ on the overlap set, together with $\kappa > 0$, make $\Sigma[Y_+]$ nonnegative and not identically zero, so Assumption 4.3 forces $\eta[Y_+] > 0$, hence both yields are nonzero and opposite. $\square$

Corollary 4.5 (Labelling). *No dynamical law selecting Y_+ over Y_- is required. Equal weights on the two histories cancel globally while each selected branch carries a nonzero asymmetry.*

Corollary 4.6 (Freeze-out). *If $\Sigma = 0$ and $\Gamma_W = 0$ for all $t \geq t_f$, then $\eta(t) = \eta(t_f)$ thereafter.*

Remark 1 (Named channel specializations). Electroweak baryon asymmetry is Theorem 4.4 with $\mathcal{O} = W\widetilde{W}$ and K built from the sphaleron window (realized in the cooling-universe companion [1]). Chiral $B-L$ asymmetry is the same theorem with $\mathcal{O} = R\widetilde{R}$ and a chiral-gravity activity kernel (catalogue row in Table 1).

5 Catalogue of interesting $\mathbb{Z}_2$ channels

Table 1 lists odd conjugations that can play the role of (3), together with a representative odd coordinate, a neutral portal in the sense of Definition 2.1, whether an early-universe activity window is structurally available, and whether Nature shows a broken yield of that type.

Column readings:

- **Active (EU)**: whether a standard early-universe processing window for the portal is known or commonly modelled (not whether it overlapped a Hysterical crossing).
- **Broken in Nature?**: observational / phenomenological status of a nonzero broken yield or an already-chiral Lagrangian, *not* a claim that Hysterical Response caused it.

Remark 2 (Three ways a row can fail to fire). Even with an odd coordinate and neutral portal, Theorem 4.4 yields nothing if (i) the mode never goes supercritical, (ii) the activity kernel K vanishes throughout the roll (no overlap), or (iii) later dynamics restores $Y \rightarrow 0$ (as in an axion settling $\bar{\theta} \rightarrow 0$). Strong CP and dark stabilizing $\mathbb{Z}_2$ are the cleanest “portal exists, broken yield absent” rows.

6 Weak parity and a possible parent Lagrangian

Status in the Standard Model. Parity is already maximally violated by the chiral electroweak Lagrangian: only left-handed fermion doublets couple to $SU(2)_L$. That is *not* a missed Hysterical branch in the sense of a still-symmetric SM waiting to tip. It is an input of the observed Lagrangian.

Symmetry / $\mathbb{Z}_2$	Odd coordinate	Neutral portal	Active (EU)	Broken in Nature?
CP	CP-odd phase Y_{CP}	$YF\tilde{F}, YG\tilde{G}, YW\tilde{W}$	Yes (windows)	Yes (CKM; η_B); strong CP tiny
C (matter $\leftrightarrow$ antimatter)	same as CP-odd charge bias	anomaly + chemical potential	Yes	Yes ($\eta_B \neq 0$ in our patch)
P / weak chirality	chiral / mirror order Y_P	LR Higgs / chiral gauge portals (BSM parent)	Model-dep.	Yes in SM Lagrangian; see §6
T	rolling odd history $\dot{Y}$	same topological portals	With CP	Yes (Kaon/B factories; tiny)
B (baryon)	CP-odd Y with B portal	$YW\tilde{W}$ + sphalerons	Yes (T_*-T_c)	Yes (η_B); B frozen today
$B-L$ / lepton	CP-odd Y	$YR\tilde{R}$ (grav. anomaly)	Conditional	Yes if leptogenesis; else unknown
Strong CP / θ	θ / axion-like Y_θ	$YG\tilde{G}$	Yes (QCD)	No ($\bar{\theta} \sim 0$)
PQ / axion misalignment	axion phase	$YG\tilde{G}$	Yes	Misalignment model-dep.; θ small
Magnetic helicity	helical CS number	Chern–Simons / chiral MHD	Yes (EW/MHD)	Unknown (not established)
GW helicity	chiral tensor amplitude	$YR\tilde{R}$	Inflationary models	Not established on CMB scales
Cosmic birefringence	axion–photon phase	$YF\tilde{F}$	Possible	Consistent with ~ 0 (contested)
CME / axial charge	axial μ_5	$E \cdot B$ / topology	Heavy-ion / early QGP	Lab hints; no proven cosmology
Affleck–Dine phase	condensate phase	B - or L -carrying AD portal	Yes (SUSY/flat directions)	Model-dep. baryogenesis candidate
Asymmetric DM	dark odd charge Y_D	dark anomaly / shared asymmetry	Model-dep.	Unknown (DM asymmetry unproven)
Dark stabilizing $\mathbb{Z}_2$	dark parity-odd field	Higgs portal / kinetic mixing	Optional	No (kept unbroken for DM)
Mirror / left–right $\mathbb{Z}_2$	mirror-odd Higgs Y_{LR}	LR / mirror portals	Model-dep.	No observed mirror twin; see §6
R-parity (if SUSY)	R-odd superfields	R-violating operators (optional)	Optional	Often assumed unbroken
Defect / winding sense	net winding N_w	odd bias of nucleation rates	Model-dep.	Unknown

Table 1: Interesting $\mathbb{Z}_2$ channels for Theorem 4.4. “Neutral portal” means Definition 2.1. No row asserts a Hysterical causation claim.

Speculation (not a theorem). One interpretive possibility, compatible with Theorem 4.4 but not implied by it, is that the “full” ultraviolet theory is $\mathbb{Z}_2$ -even under a left–right or mirror conjugation, with a parent Lagrangian schematically

$$\mathcal{L}_{\text{parent}} = \mathcal{L}_L + \mathcal{L}_R + \mathcal{L}_{\text{mix}}, \quad (13)$$

symmetric under $L \leftrightarrow R$ (or sector $\leftrightarrow$ mirror) when the odd Hysterical coordinate vanishes. A supercritical odd mode Y_P with neutral portal into the LR / mirror Higgs or gauge sector could spontaneously select a broken vacuum in which only one chiral block remains light / coupled in the infrared. On the selected branch the effective Lagrangian would look like the observed chiral SM; on the opposite branch the conjugate chirality would be isolated instead. In that reading, weak parity is “already broken” in our classical patch because we inhabit one Hysterical branch of a more symmetric parent theory—the same labelling logic as Corollary 4.5, applied to chirality rather than to η_B .

What this does and does not claim.

- **Does:** offer a structural analogy between Paper 7’s matter/antimatter labelling and a possible left/right labelling.
- **Does not:** derive the SM from an LR-symmetric parent; fix the portal coefficients; or assert that mirror fermions exist in our patch. Left–right symmetric models already exist as ordinary BSM constructions [4, 5]; the novelty here is only the Hysterical branch-selection gloss, which remains optional.

Modeling Assumption 6.1 (Parent chirality as an odd channel). Identify Y in Theorem 4.4 with a mirror-odd / LR-odd collective coordinate, take $\mathcal{O}$ to be the corresponding odd Higgs or gauge operator in (13), and let K be nonzero in an early window where both chiral blocks are still dynamical. Then (12) supplies opposite branch-conditioned chiral effective theories, without selecting a preferred global hand.

7 Logical status

Quantity / claim	Status
Master asymmetry Theorem 4.4	Theorem (Lean).
Reciprocal destabilisation 3.2	Theorem (Lean).
Catalogue Table 1	Structural / observational ledger.
Named channel recoveries	Instances (Remark 1).
Weak-parity parent Lagrangian	Speculation / Model 6.1.
Causation of any row’s broken yield	Not claimed.

8 Conclusion

Spontaneous $\mathbb{Z}_2$ breaking via Hysterical instability is the common spine of the baryon and chiral $B-L$ constructions:

$$\boxed{\text{odd supercritical } Y \implies Y_{\pm} \xrightarrow{\text{odd portal} + \text{overlap}} \eta[Y_+] = -\eta[Y_-] \neq 0.} \quad (14)$$

Table 1 records where Nature may already show such a broken yield, where portals exist without a broken yield, and where a discrete symmetry appears to be kept intact. Whether the chiral Standard Model itself is a branch-selected remnant of a symmetric parent Lagrangian is an open interpretive question, not a theorem of the response package.

References

- [1] Andrew Bruce Boyd. Asymmetric baryogenesis from electroweak hysterical instability in a cooling universe. 2026. Working mathematical formulation; electroweak cooling baryogenesis companion.
- [2] Andrew Bruce Boyd. Hysterical response in open quantum systems. 2026. Working mathematical formulation; companion response-theory paper.
- [3] Andrew Bruce Boyd. Late-universe cosmological response. 2026. Working mathematical formulation; companion late-universe paper.
- [4] R. N. Mohapatra and Jogesh C. Pati. A natural left-right symmetry. *Phys. Rev. D*, 11:2558–2561, 1975.
- [5] Goran Senjanović and Rabindra N. Mohapatra. Exact left-right symmetry and spontaneous violation of parity. *Phys. Rev. D*, 12:1502–1505, 1975.